Iron Valley (Mountain / Mining) Map Design

Executive Summary

Iron Valley is a harsh mountainous mining battlefield designed for ground-focused warfare, logistics pressure, and brutal chokepoint engagements. The terrain features steep canyon walls, collapsed tunnels, active mining machinery, ore conveyors, and industrial refineries. The map supports Obsidian Protocol's autonomous decision -making, sensor-driven tactics, comms-relay warfare, and logistics-heavy gameplay loops. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of terrain. Key features include narrow canyon roads, ridge -top sensor towers, mining pits, ore-processing facilities, and critical supply depots.

Design Goals & Gameplay

Ground Warfare & Scale

Iron Valley emphasizes infantry and armor combat. The valley floor provides maneuver space. while canyon walls restrict movement and create natural ambush funnels. Mining roads and industrial yards allow heavy vehicles but limit air support due to dust. smoke. and machinery interference.

Autonomy & Flanking

Multiple routes — canyon passes, collapsed tunnels, conveyor-belt walkways, and ridge-top paths — create nonlinear movement. Autonomous AI can choose ambush points, retreat paths, or flanking routes without micromanagement.

Sensor & Information Warfare

Dust storms, machinery vibration, and ore -processing heat distort radar and thermal sensors. Ridge-top relay towers provide clean sensor coverage. Destroying relays creates fog-of-war pockets and comms dead zones.

Logistics & Resources

Mining trucks, ore containers, and fuel depots serve as logistics nodes. Long supply lines through narrow roads force players to protect convoys. Controlling the refinery and fuel depot grants major resupply advantages.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces strategic tradeoffs: heavy armor through the canyon, or fast recon along ridge paths? Blue deploys from the west, Red from the east, with a neutral forward depot near the central mine.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Comfortable viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² battlefield (1:100 scale). Fits on a conference table.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; design assumes minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Location	Coordinates	Size	Description
West Base	(0,0)	100×100 m	Elevated staging yard with mining vehicles
East Base	########	100×100 m	Cliff-side command post with comms tower
Central Mine	-500,400	180×180 m	Excavation pit, conveyor belts, crushers
Ore Refinery	-800,250	90×90 m	Smelters, cooling towers, heavy pipes
Fuel Depot	-250,800	60×60 m	Fuel tanks, supply crates, truck bays
North Pass	(400,1000	—	Narrow canyon, blocked LOS
South Tunnel	(300,0 -> 4	—	Collapsed mine shaft, close-quarters combat
River & Br	(0,500 -> 4	—	Two crossings at (150,600) and (350,500)
Peak A	-600,100	+300 m	Sensor vantage point
Peak B	-100,600	+250 m	Ridge-top comms relay
Neutral Fo	-500,950	—	Drone resupply zone

Chokepoints & Sightlines

North Pass

A tight canyon corridor with steep walls; LOS blocked except from ridge-top units.

South Tunnel

Collapsed mining tunnel; extremely close-range engagements.

Bridges

Two river crossings; controlling them prevents enemy flanking.

Sightlines

Peak A provides long-range sensor coverage (~300 m). Deep valleys create fog-of-war pockets. Ridge-top comms tower at East Base covers most of the map.

Deployment Zones

Blue Deployment: West Base + rear rally point.
Red Deployment: East Base + rear rally point.
Neutral Forward Zone: Airfield at (500,950) for drone droos.

Assets & Environment

Terrain

Rocky cliffs, slag piles, mine shafts, conveyor belts, ore pits.

Vegetation

Sparse pines, shrubs, grass patches (billboards).

Buildings & Props

Industrial: Crushers, smelters, ore silos, pipelines.
Military: Watchtowers, antennas, radar dishes.
Logistics: Mining trucks, crates, barrels, forklifts.
Roads/Bridges: Canyon roads, metal truss bridges.

Effects

Dust storms, smoke plumes, furnace glow, conveyor sparks.

Audio

Wind gusts, machinery hum, ore crushers, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Mountain	3	200k	600k
Slopes/Hil	4	80k	320k
Buildings	15	50k	750k
Bridges/Ra	4	30k	120k
Vegetation	150	2k	300k
Props	50	10k	500k
Total	—	—	**2.59M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, GPU-friendly lightmaps.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

Milestone	Timeframe	Deliverables
1. Prototy	Month 1	Block-out terrain, bases, mine pit, AR anchor test
2. Core Sy	Month 2	NavMesh. pathfinding. autonomy behaviors
3. AR Inte	Month 3	Anchors. plane detection. basic interactivity
4. Content	Month 4–!	Final models, textures, lighting, audio
5. Gameol	Month 6–:	Objectives, balance, VFX (dust, sparks)
6. Testing	Month 8	Multiplayer stress test, sensor validation

Testing Plan

Autonomy

Verify AI intent logic (attack, hold, retreat).

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, cover usage, deployment budget enforcement, anchor drift, sensor accuracy.